

Impact of roots on ground water quality

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Abstract

Preferential flow is perhaps the major chemical transport process influencing the rapid and typically unexpected movement of agricultural chemicals to ground water. Plant roots are a major contributor to preferential flow mechanics as they form spatial voids which can be used as preferential flow pathways. Chemical transport of atrazine, deethylatrazine, and bromide solutions concentrations under tilled and no-tilled corn fields was evaluated below the active root zone. Additionally, the impact of roots on flow pathways was visualized using a soluble dye (Brilliant Blue FCF). Pictures of the dye-stained pattern were subsequently digitized to determine the cross-sectional area used for transport as a function of depth. Bromide transit times through the field soils were not influenced by tillage practice, whereas atrazine transport was strongly influenced by tillage practice. Under no-till field conditions, atrazine was rarely detected but deethylatrazine concentrations were greater than those observed under tilled field conditions. Visual observation indicated that the dye under no-tillage was more predominant in the corn row, indicative of transport through void root channels. No-tillage practices may decrease the likelihood of ground water contamination through leaching due to the formation of stable root channels where an organic carbon source and microbial population are preferentially located to degrade pesticides.

Introduction

Most research efforts on minimizing detrimental environmental effects of agricultural chemicals have focused on a specific component of the environment, i.e., soil, water or air. However, each of these components are intimately linked and together influence ground water quality. In addition, although techniques have been developed for evaluating chemical fluxes from runoff and vapor losses to the atmosphere, there is no method for accurately measuring and evaluating leachate fluxes on a field scale. The complex heterogeneous nature of the soil matrix gives rise to highly variable water fluxes, making it nearly impossible to quantify a water and solute flux below the root zone. Recently, preferential flow and transport have been identified as major contributors to the variable nature of water and solute fluxes in the field (Gish et al., 1991; Richards and Steenhuis, 1988) of which plant roots

can be a major contributor (Gish and Jury, 1983). The development of environmentally safe and cost effective agricultural management strategies is ultimately linked to our ability to detect and measure leachate fluxes on a field-scale.

A generic term used to describe the non-homogeneous movement of water through soil is 'preferential flow'. During the last decade, it has been gradually recognized that preferential flow is the most significant field-scale leaching process influencing the pollution potential of agricultural chemicals. In general, there are three types of preferential flow mechanisms that have been identified: (1) macropore flow (Bouma, 1981; Edwards et al., 1988; Germann and Beven, 1981); (2) finger flow (Baker and Hillel, 1990; Glass et al., 1988; Hill and Parlange, 1972); and (3) funnel flow (Ju and Kung, 1993; Ju et al., 1996; Kung 1990a,b). Macropore flow describes the movement of water through continuous voids created by earthworms, decayed roots, and soil fractures. Typically, finger flow occurs when a uniform wetting front moves

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into a very coarse and dry soil, after which the water breaks up into discrete column-like flow paths. Funnel flow occurs along any inclined textural boundary with an abrupt change in pore size distribution which can congregate water into particular flow pathways. All three types of preferential flow can occur simultaneously in a field. Examples of preferential transport in the field include Kladvik et al. (1991), who found that, after only 1 cm of net infiltration, high concentrations of pesticides were detected in water samples collected from tile drains buried at 75 cm in a silty loam soil. One pore volume for this latter soil was approximately 30 cm of water. Quisenberry et al. (1991) showed that, in structured loamy soils, water and solutes moved down uniformly in the top of the Ap horizon. However, in the lower horizon, more than 75% of the solutes moved through less than 25% of the soil matrix. In a layered soil profile, Roth et al. (1991) found that a uniformly-applied chemical pulse was split into a slow and fast front. The fast front reached the 2.3-m depth after only 3.1 cm of natural precipitation. On the other hand, the slow moving front moved to only the 0.84-m depth after 85.3 cm of net infiltration. These latter investigators also estimated that only 42% of their tracer was recovered from the slowly moving front, suggesting that 58% of the applied tracer moved deeper into the soil through preferential flow pathways.

The major drawback for evaluating the scope and magnitude of preferential flow on a field-scale is our inability to detect and measure it. Historically, soil cores were removed and quantitatively analyzed as a function of depth for various agricultural chemicals. This sampling protocol was developed more than 50 years ago and is based on the assumption that: (1) water and solutes move through the entire homogeneous matrix; and that (2) the soil and soil solution are in instantaneous equilibrium. Unfortunately, the spatial and temporal complexities associated with preferential flow violate the assumptions of homogeneous, equilibrium transport. For example, Wyman et al. (1985, 1986) collected 48, 3.6-m long soil cores four times each growing season. Each soil core was segmented into 30-cm increments, where each segment was homogenized and subsampled for aldicarb. As no aldicarb residues were detected below 2.4 m in all samples throughout the three-year period, these authors concluded that aldicarb was completely degraded and would not pose a threat to groundwater quality. However, with the same chemical application rate, soil, time, and irrigation scheme, Brasino (1986) found peak aldicarb concentrations in three ground water

monitoring wells (6-m depth) ranging from 27 to 76 ppb. Furthermore, Gish et al. (1996) observed that even after atrazine solution and soil core concentrations were each normalized to a concentration per soil volume, the solution phase concentration was at least 20 times greater than that observed in the soil fraction. These field studies suggest that the transport of agricultural chemicals in soil, in many cases, is not well described and that preferential transport is a common flow process. Recently, the US Environmental Protection Agency (EPA) revised their standards for water quality investigations. Although soil cores are used to characterize the site, only solution phase concentrations will be used to evaluate field-scale leaching processes. In addition, these studies indicate that chemical leaching processes cannot be determined from soil core data alone.

Many examples of preferential transport were summarized by Helling and Gish (1991) and Luxmoore (1991). However, the mechanisms, scope, and magnitudes of preferential flow under field conditions have not been fully characterized, especially in the lower part of the vadose zone (McCoy et al., 1994). The morphological features of preferential flow pathways have been visualized and identified by using soluble stains. Methylene blue, used to stain macropores for visual observation and for fractal analysis (Hatano et al., 1992), is strongly adsorbed, as such it would represent the leaching behavior of strongly adsorbed pesticides. An iodine-coloring method has been developed for depicting the flow pathways of more mobile compounds as nitrate (van Ommen et al., 1988). Unfortunately, both iodine and methylene blue sharply reduce microbial activity in soil. Recently, an alternative dye, Brilliant Blue FCF, has been used successfully in evaluating preferential water flow patterns (Flury et al., 1994).

Unlike preferential flow pathways that are induced by tillage operations, preferential flow pathways formed by root activity may have a positive impact on ground water quality. Water flow through void root channels can be rapid, but organic coatings on old root channels could facilitate biological degradation of preferentially moving agricultural chemicals. As a result, this study was conducted to compare and determine the impact of void roots channels and man-induced flow channels, created by tillage operations, on field-scale behavior of bromide, atrazine and deethylatrazine.

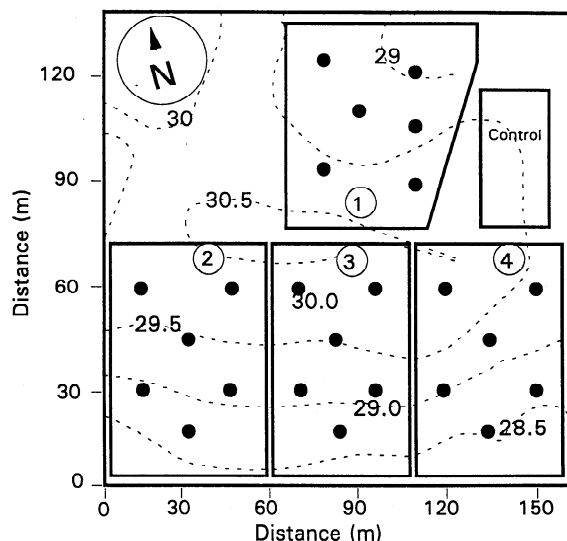


Figure 1. Topographical representation of the field site (meters above mean sea level). Fields 1 and 2 were under no-tillage practices, with fields 3 and 4 under tilled practices. The dots represent the spatial location of the solution lysimeters.

Materials and methods

The research site is located at the Maryland State Extension Farm in Upper Marlboro, MD. The site consists of four fields, each about 0.25-ha fields that are topographically bounded. The topographically bounded nature of the field minimizes hydrologic uncertainty relative to surface water inputs to the research site. Prior to 1989, the fields were in pasture grass and had no history of herbicide application. In 1989, the entire research site was tilled and berms established around each field. In 1990, fields 3 and 4 were plowed while fields 1 and 2 were dedicated to no-till management (Figure 1). All fields were under continuous corn production with rye planted each winter as a winter cover crop. Except for the use of the no-till drill, the no-tillage fields remained undisturbed until the spring of 1996. Fields 2 and 4 had received herbicides in an encapsulated matrix, while fields 1 and 3 had received only commercially available herbicides (Gish et al., 1995).

In 1990, six sets of suction lysimeters were installed in each field between 1.5 and 1.8 m (Figure 1), for evaluating solution-phase concentrations exiting the root zone. Solution samplers were used in this study because of the non-destructive nature of the sampling procedure and they allow the same field

area to be monitored over time. Solution sampling occurred after each rain event ≥ 1 cm; 24 hr after a rain event, suction (-0.12 kPa) was applied to each lysimeter. On the following day, solution samples were removed, and stored in glass bottles at 4°C until analyzed for bromide, atrazine [6-chloro-N-ethyl-N'-(1-methylethyl)-1,3,5-triazine-3,4-diamine] and deethylatrazine [6-chloro-N-(1-methylethyl)-1,3,5-atrazine-2,4-diamine]. During the 2-yr period, the average sample volume was 0.094 L for the tilled field lysimeters and 0.092 L for the no-tillage lysimeters.

Chemical applications consisted of an annual application of atrazine. On June 5, 1991 atrazine ($1.7\text{ kg a.i. ha}^{-1}$) and bromide (80 kg ha^{-1}) were co-applied to the field-site as a surface broadcast spray. Bromide was used as a solute tracer, to monitor water movement through the soil profile and to determine the impact of tillage on chemical transport (see Gish et al., 1995 for details). Atrazine ($1.7\text{ kg a.i. ha}^{-1}$) was subsequently applied once annually between May 28 and June 5, for each year up until 1996. In 1996, the fields were prepared for planting in the identical manner they had been the prior 7 yr. Two weeks before chemical application the rye was cut. In the no-tillage fields the rye was left on the soil surface while on the tilled fields the rye was incorporated into the soil and the fields subsequently cultipacked. Unlike in prior years, the fields were left fallow. After two weeks, a 0.5% solution of Brilliant Blue was sprinkler-applied as a simulated 7-cm rain event. The solutions were applied as fast as possible without runoff, though microponding was allowed. As a result, it took about 1.3 h to apply the solutions on the no-till plots and about 1.8 h under tilled conditions. A 7-cm rain event over a 1.5–2 h period is common to this region, having a return period of about 2 yr. Three days after the dye was applied, a large pit ($2.2 \times 3.2\text{ m}$) was excavated in both fields 1 and 3 in an attempt to visualize Brilliant Blue flow pathways, and relating the dye patterns to observed atrazine and bromide concentrations acquired from the solution lysimeters.

Bromide was determined in 5-mL aliquots from the lysimeter samples, passing the sample through a $0.45\text{-}\mu\text{m}$ filter, and quantifying by ion chromatography. The ion chromatograph utilized a mobile phase of 1.0 mM potassium hydrogen phthalate titrated to pH 6.5 with lithium hydroxide; flow rate was 1.2 mL min^{-1} ; and the sample injection volume was $100\text{ }\mu\text{L}$. Detection limits for Br^- were 0.05 mg L^{-1} .

Atrazine and deethylatrazine (DEET) were quantitatively analyzed. Solutions were passed through

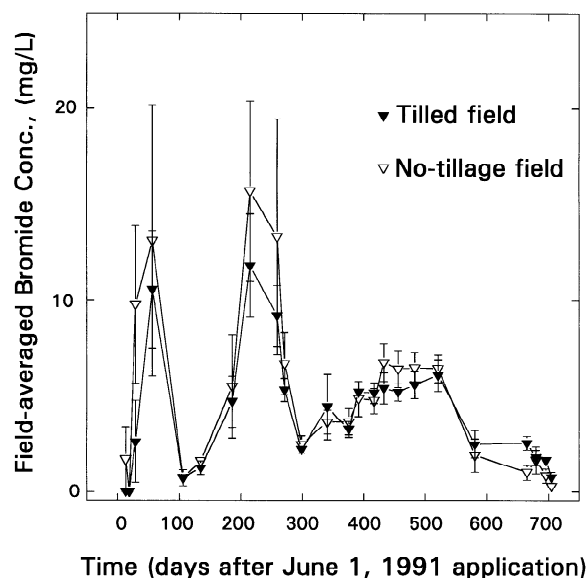


Figure 2. Averaged bromide leachate concentrations below the root zone of field-site as a function of tillage practice. Averaged concentrations were calculated from combining data from the 1.5 and 1.8-m depths. Error bars signify ± 1 standard error of the mean.

Waters C-18 Sep-Pak¹ (Waters Associates) solid-phase extraction cartridges (Nash, 1990), then eluted with ethyl acetate. Trifluralin [2,6-dinitro-N,N-dipropyl-4-(trifluoromethyl) benzenamine] was added to the eluant as an internal standard for gas chromatography (GC) quantification. Gas chromatograph operating conditions utilized a 30-m $\times$ 0.32-mm i.d. glass capillary column coated with 0.26- μ m SPB-5 (Supelco, Bellefonte, PA); injector temperature, 200 °C; oven temperature, 150 °C; N-P detector temperature, 220 °C; and He carrier gas flow rate of 2.5 mL min⁻¹. Atrazine recoveries from spiked solution samples ranged from 97–100%. Detection limits were 0.01 μ g L⁻¹ for atrazine and 0.05 μ g L⁻¹ for DEET.

Results and discussion

Lysimeter data

Bromide leachate concentrations from the solution lysimeters clearly show that preferential Br⁻ transport occurred under both tillage systems (Figure 2). Since no Br⁻ had been applied to the fields prior to 1991, these Br⁻ concentrations are assumed to be a result of that

applied. Detailed analysis of the Br⁻ data also indicates that there are actually two peaks, with a long diffuse bromide tail (for a more in-depth analysis of the curves see Gish et al., 1995). Multiple peaks from solute tracers in structured or fractured media is not uncommon (Nordquist et al., 1992; Tsang et al., 1991). The first peak is a result of preferential flow and transport, while the second peak is due to convective-diffusive transport. The diffuse Br⁻ tail is a consequence of bromide diffusing out of dead end pores and/or chemical transport through the small pores that are significantly smaller than the average pore diameter (i.e. transport within soil peds).

The first Br⁻ peak had a maximum concentration at about 60 d after application under both tillage regimes. Since, a total of 21.9 cm of water had fallen onto this site during the first 60 d, the Br⁻ front should have reached a maximum depth of about 0.6 m (assuming homogenous transport and no evapotranspiration), a full meter shallower than the depth of the suction lysimeters where the samples were taken. Preferential transport on sandy soil (Glass et al., 1988; Kung, 1990a) as well as on structured no-tillage (Gish et al., 1991; Isensee et al., 1990) has been reported, but the similarity in maximum concentrations, or the time in which the maximum concentrations appeared in the solution samplers for the tilled and no-tillage soils of our study was not expected.

Bromide leaching behavior was similar between the two tillage practices, over the 2-yr study. However, certain significant differences occurred: early (29 d after application) in the preferential flow peak, the field-average Br⁻ concentration was 9.8 mg L⁻¹ for no-tillage (standard error of the mean ± 4.1) but only 2.6 mg L⁻¹ (standard error of the mean ± 2.1) in the tilled fields. Preferential flow pathways resulting from roots and earthworm activity have resulted in increased infiltration rates and rapid solute transport (Edwards et al., 1990, 1993). Consequently, although preferential flow occurred under both tillage regimes, it may have been more significant under the no-tillage practice. Unfortunately, analysis of the Br⁻ concentrations themselves cannot distinguish the type of preferential flow that was most significant for solute transport (i.e. spatial voids, fingering or funnel flow). The second bromide peak that occurs (between 200 and 270 d) was expected if estimated net water inputs are calculated from pan evaporation and rainfall measurements and used as a water input for the classical convection-dispersion transport model (CXFIT; Parker and van Genuchten, 1984). The long Br⁻ tail that takes about

¹ Trade names included for the benefit of the reader and imply no endorsement or preferential treatment of the product listed by the US Department of Agriculture.

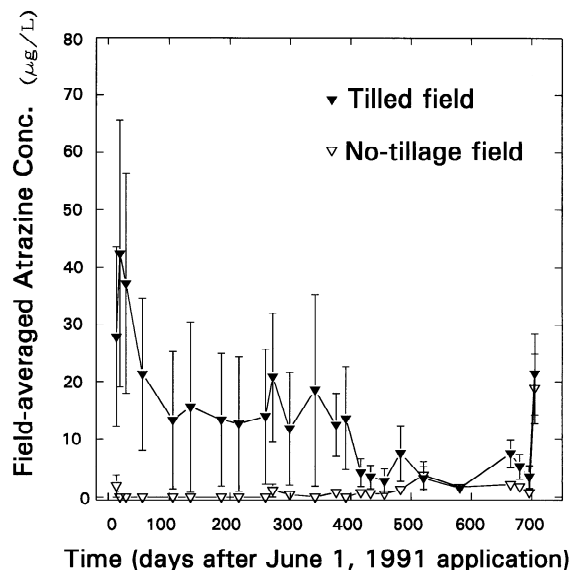


Figure 3. Averaged atrazine leachate concentrations below the root zone of the field-site as a function of tillage practice. Averaged concentrations were calculated from combining data from the 1.5 and 1.8-m depths. Error bars signify ± 1 standard error of the mean.

1 yr to complete is a result of the reduced water velocities occurring in the very small pores of the soil matrix. In soil columns, this type of tailing is evidenced by asymmetric solute breakthrough curves. Owens and Edwards (1992) observed similar Br^- tailing in a 1.1-ha watershed that had received 168 kg ha^{-1} of the tracer.

Atrazine concentrations during the same 2-yr period are shown in Figure 3. From depths below the active root zone (1.5 and 1.8 m), 53% of the solution samples located on the tilled field contained atrazine, compared to only 28% of the solution samples collected under the no-tilled field. Initially, these data appeared incongruous relative to the Br^- data. Gish et al. (1995) proposed that under no-tillage practices, preferential transport was likely a result of spatial voids created by root channels and worm holes, whereas the preferential transport in tilled sandy soil was likely a result of unstable wetting fronts and fingering. To support this hypothesis, mean DEET concentrations in solution samples showed only a slightly larger concentration ($0.52 \mu\text{g L}^{-1}$) under no-tillage compared with tilled-field results ($0.39 \mu\text{g L}^{-1}$). However, if DEET concentrations are compared for two more years, results change dramatically: $1.29 \mu\text{g L}^{-1}$ for no-tillage (with a standard error of the mean, 0.18), but only $0.34 \mu\text{g L}^{-1}$ for the tilled fields. Figure 4 compares averaged DEET concentrations from the tilled and no-tillage fields for

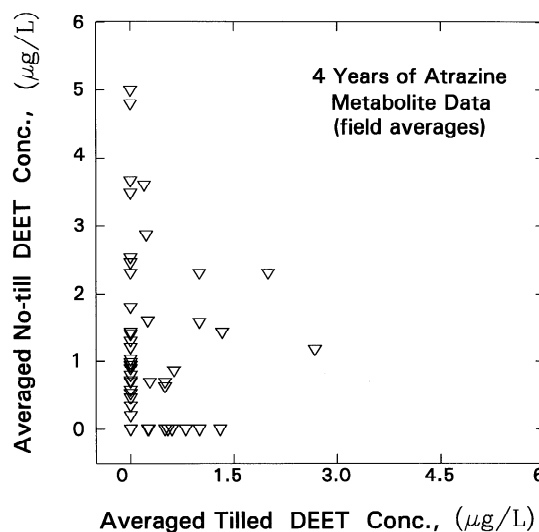


Figure 4. Comparison of the atrazine metabolite, deethylatrazine (DEET) under tilled and no-tillage fields that had received commercially formulated herbicides. Each triangle represents a tillage treatment average DEET concentration for a specific sampling date.

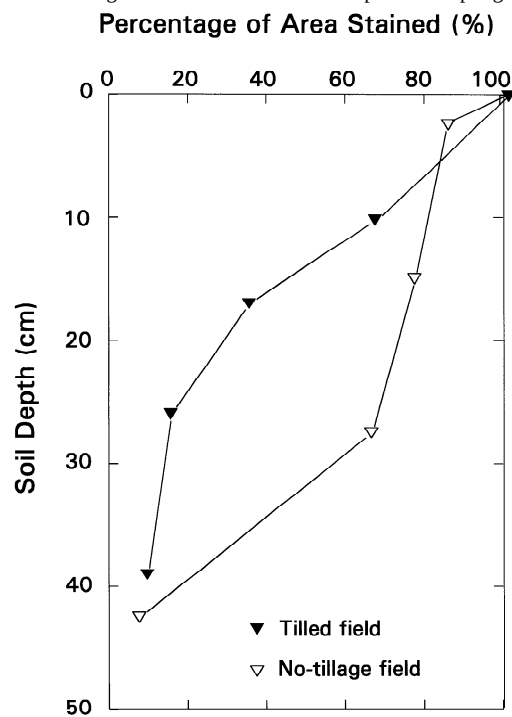


Figure 5. Percentage of area stained by Brilliant Blue FCF as a function of depth in tilled and no-tillage fields.

each observation time (each dot represents a sampling date). Values scattered along the y-axis indicate that field-averaged DEET concentrations increased under

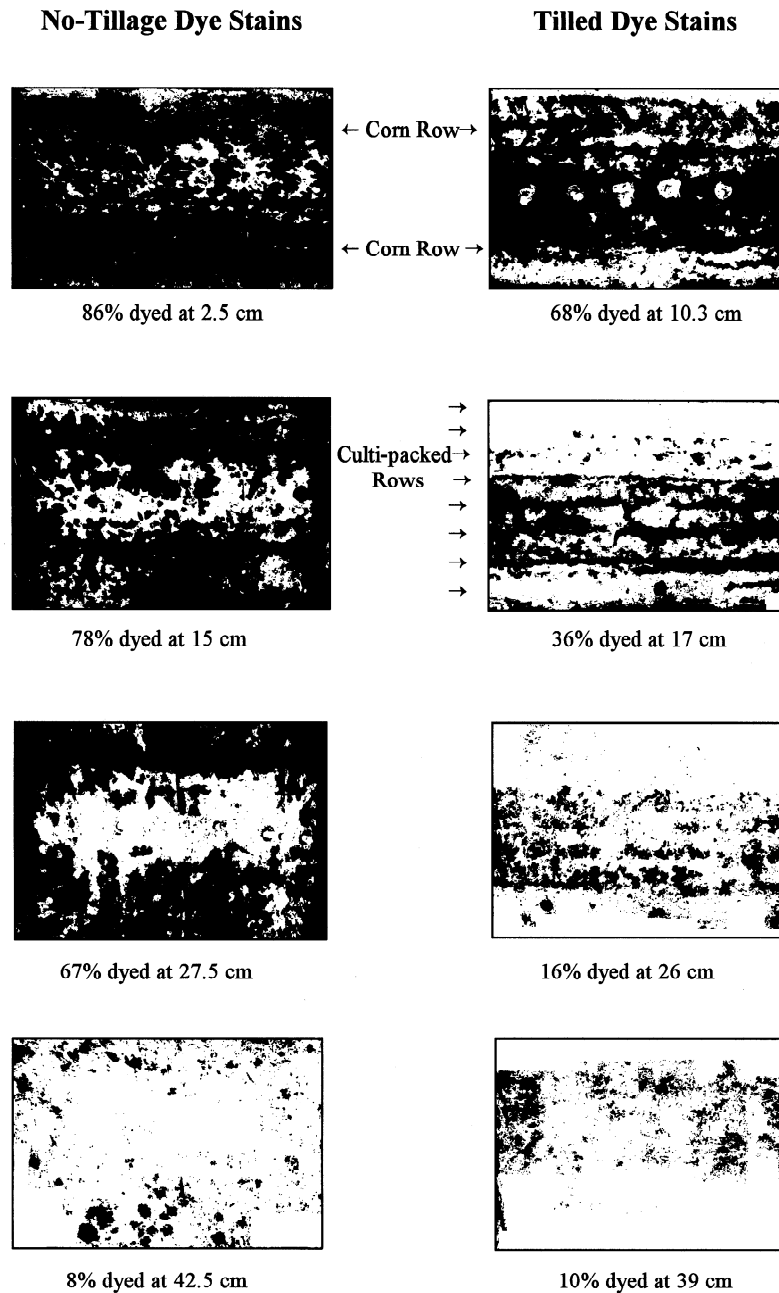


Figure 6. Spatial location of dyed areas (shown in black) in tilled and no-tillage fields. Views represent horizontal cut at depths indicated.

no-tillage, while at the same time there was essentially no detected of DEET under the tilled field. As a result, it appears that the formation of at least this atrazine metabolite increased significantly with time under no-tillage, but remained nearly the same under the tilled fields.

Dye patterns

The fractional area of the soil used for flow as a function of depth is depicted in Figure 5. Simple piston flow theory predicts that the dye front should stop at about a 20-cm depth, but some preferential transport was apparent based on deeper instances of soil dyeing.

Regardless of tillage practice, between 8 and 10% of the soil area is still being used for chemical transport at about 40 cm. However, the relationship between the stained flow areas and depth does vary with tillage. At around 16 cm, twice as much soil area is stained under no-tillage than with the tilled soil. Additionally, at around 27 cm, four times as much area is stained under no-tillage than with the tilled soil. As a result, even though preferential chemical transport occurs under both tillage regimes, it appears to be more dominant under no-tillage.

The spatial location of the flow pathways as a function of depth are shown in Figure 6. In the no-tillage plot, the dye is concentrated in the corn rows away from the solution lysimeters, especially near the surface. This is due, in part, because these solution samplers extend above the soil surface, and as such could not be approached by the no-till chisel without damaging the suction lysimeters. Macropores were also frequently observed in the no-tillage pits, some which were used for dye transport. Since the dyed areas are linked to the corn rows, where old root and worm channels were plentiful, any solute moving through these channels will also be subjected to enhanced microbial activity resulting from organic coatings. These organic coatings or organic cutans, are the result of root decomposition and worm castings (Greenland, 1979; Pivetz and Steenhuis, 1995). As a result, a herbicide moving preferentially through the old root channels would also be subject to adsorption onto organic coatings and enhanced microbial activity (relative to the bulk soil) along the root channel.

The location of the flow pathways in the tilled soil are quite different than those observed under no-tillage. In the tilled plot, the preferential flow pathways are associated with the cultipacking operation and not with the location of the corn rows. As a result, the solutes moving along these flow pathways may not be subjected to the same microbial activity that exists within the root zone. This may be why Stearman et al. (1989) observed a decrease in herbicide activity in tilled soils and why less metabolite was detected below the tilled root zone of this field site.

Summary

Transport of atrazine, DEET, and bromide below the active root zone of tilled and no-tillage fields was monitored and evaluated. Over a 2-yr period, bromide leaching from this sandy soil was nearly identi-

cal under both tillage regimes and indicated significant preferential transport. Unlike bromide, atrazine transport was quite different with less parent compound and more metabolite under no-tillage than under the tilled fields. Utilizing a readily soluble food dye, the flow pathways were stained and tillage systems compared. Although preferential transport was common to both tillage operations, the preferential transport under no-tillage is linked to transport in old root channels. As such, chemicals that move through those void root channels may also be moving through a zone where conditions are most conducive not only to preferential transport but also to enhanced pesticide degradation: higher organic carbon content, elevated microbial populations, and higher oxygen content than in the bulk soil. On the other hand, the flow pathways visualized under the tilled field was a result of tillage operation. As a result, there are hydrologic settings where no-tillage operations will facilitate reduced runoff, and enhance biological degradation of surface applied pesticides, and as such may be a best management practice for conserving ground water quality.

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